University of Chittagong
7th Semester B.Sc. Engineering Examination 2024
Course Code: CSE 713, Title: Artificial Intelligence
Full Marks: 54, Time: 4 Hours
[Answer any three questions from each section]

Section A

1.a) Define AI. Point out a few advancements of AI recently. 1

b) Differentiate between AI and Machine Learning. 1.5

c) Evaluate the performance of Large Language Models (LLMs) in the context of the Turing Test. 2.5

i) How do LLMs exhibit human-like conversational abilities - such as fluency, contextual awareness, and reasoning—that correspond with the objectives of the Turing Test for evaluating machine intelligence?

ii) Examine the constraints of large language models, encompassing insufficient real-world grounding, the occurrence of hallucinations, the absence of stable memory or beliefs, and challenges in sustaining long-horizon dialogues. Explain why these limitations may prevent an LLM from reliably passing the Turing Test in extended, diverse, or adversarial interactions.

d) Investigate an Autonomous Agricultural Field Robot used in Bangladesh for monitoring crops, detecting pests, and spraying fertilizer. 4

(i) What are the sensory inputs (percepts) used by this robot?
(ii) Characterize its operating environment using standard AI environment properties.
(iii) What actions are available to this robot during field operations?
(iv) How can the performance of this robot be evaluated?
(v) What type of agent architecture is most appropriate for this robot, and why?

2.a) Consider the following search problem. The states are A, B, C, D, E, F, H, J, K, and G (goal state). All roads are bidirectional.

Table 1: Road Connections and Step Costs

From	To	Cost
A	B	3
A	C	4
A	D	7
B	E	6
B	F	5
C	F	11
C	H	4
D	H	2
D	J	10
E	G	7
E	H	3
F	G	9
F	J	4
H	G	5
H	K	6
J	K	3
K	G	4

Table 2: Heuristic Estimates h(n) to Goal G

Node	h(n)
A	11
B	10
C	7
D	6
E	6
F	6
H	4
J	5
K	3
G	0

ABEG
ACHG
2. ABJKG
ACHG

i) Using the information in Table 1, draw the state-space graph, clearly labelling all nodes and edge costs. Assume the initial state is A and the goal state is G. 1

ii) Using the step costs from Table 1 and heuristic values from Table 2 where appropriate, show how each of the following search strategies constructs a search tree to find a path from A to G: 5

1. Uniform-Cost Search (UCS)
2. Greedy Best-First Search (using h(n))
3. A* Search, using $f(n) = g(n) + h(n)$
4. Iterative Deepening Depth-First Search (IDDFS)

For each algorithm:

- At every step, indicate which node is being expanded, and
- Show the current contents of the fringe (open list / frontier or stack), in order of selection.
- For IDDFS, clearly indicate the depth limit used in each iteration and the order of node expansions at that limit.

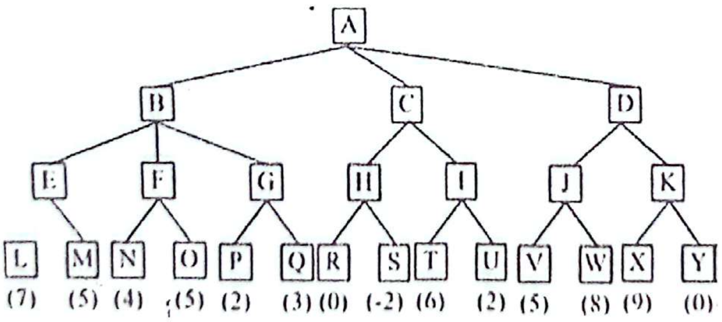
Assume ties are broken alphabetically.

b) Give the initial state, goal test, operators, and path cost function for each of the following. 3

i) Robot Vacuum Cleaning Problem (Indoor Environment): A robot vacuum operates in a house with multiple rooms. Some rooms are dirty, some are clean, and there may be static obstacles (furniture). The robot must clean all dirty rooms and try to minimize total energy usage. Briefly comment on whether this problem is more suitable for uninformed search (e.g., BFS, UCS) or informed search (e.g., A*), and justify your answer in 2–3 sentences.

ii) Water Jug Problem (5L and 3L Jugs): You are given a 5-liter jug and a 3-liter jug with no measurement markings. You have access to an unlimited water source and a drain. The goal is to obtain exactly 4 liters of water in one of the jugs. Is the state space for this problem finite or infinite under your model? Justify.

3.a) Consider the following game tree in which static scores are all from the first player's point of view: 5



i) Explain why the Min–Max algorithm becomes inefficient for large game trees. Discuss at least two limitations of pure Min–Max in adversarial search.

ii) Apply the Alpha–Beta Pruning algorithm to the tree shown above. Show clearly:
- the α and β values at every MAX and MIN node as they get updated,
- which branches are pruned,
- and the final Min–Max value returned to the root.

iii) Using your analysis, explain how Alpha–Beta pruning addresses the limitations of Min–Max in terms of:

1. time complexity
2. unnecessary node expansion
3. improved decision-making under resource limits

b) A delivery drone company in Bangladesh is attempting to optimize the flight route for a drone that must pick up packages from several rural drop-off points and return to its base. The goal is to minimize the total travel distance, but the terrain includes obstacles, shifting wind patterns, and local elevation changes that create a complex, uneven search landscape. Using this scenario: 4

i) What distinguishes deterministic optimization algorithms from non-deterministic counterparts? Do you consider hill-climbing deterministic in this context? Give a reason.

ii) Identify three limitations of hill-climbing when applied to optimizing the drone's route.

iii) Do you regard Simulated Annealing (SA) as a non-deterministic algorithm? Briefly explain how SA helps overcome the limitations of hill-climbing in this drone-route problem.

4.a) In ER (Evidential Reasoning), how a high-level attribute is assessed through associated lower-level attributes? 1

b) What is the combined degree of belief in the original ER approach? 1

c) Consider the following sentences:

i) If a perfect square is divisible by a prime P, then it is also divisible by square of P,

ii) Every perfect square is divisible by some prime,

iii) 36 is a perfect square,

iv) Does there exist a prime q such that square of q divide 36?

A) First translate the sentences above into FOL Sentences. 2

B) Prove (iv) by using Universal Elimination, Existential Elimination, Modus Ponens, and Existential Introduction. 1.5

C) Convert the FOL sentences into Clausal Form by using Canonical Form (CF). 1.5

D) Apply the Resolution method to prove (iv). 2

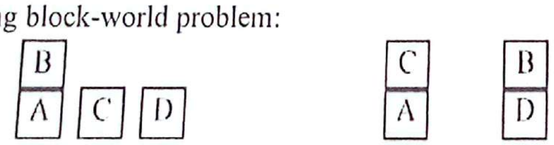
Section B

5.a) What is STRIPS, and how are actions represented using the STRIPS action schema? Develop a planning problem using STRIPS by defining one action with clearly stated: 4

- Preconditions
- Add-effects
- Delete-effects

Use the domain of a Household Robot Assistant, for example actions such as cleaning a room, turning on appliances, or fetching objects.

b) Consider the following block-world problem: 5



Start:	Goal:
ON (B,A)^	ON(C,A) ^
ONTABLE(A) ^	ON(B,D) ^
ONTABLE(C) ^	ONTABLE(A) ^
ONTABLE(D) ^	ONTABLE(D)
ARMEMPTY	

Using Partial-Order Planning (POP):

1. Construct the minimal POP plan,

2. List all causal links and ordering constraints,

3. Explain how POP avoids backtracking compared to total-order planning.

6.a) Define forward and backward chaining. What are the factors that determine whether it is better to reason by using forward or backward chaining? 2

b) What is conflict resolution? Discuss any two of the approaches which are used to solve conflict resolution. 1.5

c) Draw the architecture of a rule-based system. Following are the rules and initial facts in the rule base and database of facts of a rule-based expert system. 5.5

R1: IF A and C THEN E	Initial facts in the database
R2: IF C and D THEN F	A, B (both are true)
R3: IF B and E THEN F	Objective:
R4: IF B THEN C	Prove hypothesis G (goal)
R5: IF F THEN G	

Prove the goal (G) by applying both Forward Chaining and Backward Chaining inference procedures. You need to show the sequence of firing the rules and also need to draw the search tree, in which propagation of truth for FC and backtracking of goals for BC to be demonstrated diagrammatically.

7.a) A robot vacuum suddenly stops working (Evidence E). The possible hypotheses are: 4

- H_1: Battery is low
- H_2: Dust container is full
- H_3: Motor is overheated

From past maintenance logs:

- $P(H_1) = 0.5$, $P(H_2) = 0.3$, $P(H_3) = 0.2$
- $P(E \mid H_1) = 0.7$, $P(E \mid H_2) = 0.8$, $P(E \mid H_3) = 0.9$

Using Bayes' Theorem:

1. Compute the total probability of the evidence P(E).
2. Calculate the posterior probabilities $P(H_1|E)$, $P(H_2|E)$, and $P(H_3|E)$.
3. Identify which hypothesis is most likely the cause of the malfunction.
4. Justify your final decision.

b) What are the challenges in Bayesian decision theory? 2

c) What do you mean by evolutionary method? Explain with an example. 3

8.a) What is learning? Describe the inductive learning method with necessary example. 3

b) Compare artificial and biological networks. What aspects of biological networks are not mimicked by artificial ones? What aspects are similar? 3

c) Write short notes on the following: 3

i) Fuzzy logic

ii) Uncertainty

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CSE 713: Artificial Intelligence
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Answer three questions from each section. Read the text of each question carefully before solving it. Figures in the right-hand margin indicate full marks.

Section A

1. (a) Define Artificial Intelligence (AI). How can the connections among different forms of AI be clarified through an examination of their generative traits, capabilities, and functional attributes? [2]

   (b) Evaluate the performance of ChatGPT in the context of the Turing Test. In your response, address the following: [3]

   i. In what ways does ChatGPT correspond with the goals of the Turing Test regarding the emulation of human-like conversation?

   ii. Discuss the limitations of ChatGPT that would prevent it from passing the Turing Test in more complex or prolonged interactions.

   (c) Investigate the most recent iterations of the Mars Rovers, specifically Curiosity, which was launched in 2012, and Perseverance, which followed in 2021. [4]

   i. What are the sensory inputs (percepts) for these agents?

   ii. Characterize their operating environment.

   iii. What actions are available to these agents?

   iv. How can one evaluate the performance of these agents?

   v. What type of agent architecture do you consider to be the most appropriate for these agents, and what are your reasons for this assessment?

2. (a) Give the initial state, goal test, operators, and path cost function for each of the following. [2]

   i. In the travelling salesperson problem (TSP) there is a map involving N cities some of which are connected by roads. The aim is to find the shortest tour that starts from a city, visits all the cities exactly once and comes back to the starting city.

   ii. Missionaries and Cannibals problem: 3 missionaries and 3 Cannibals are one side of the river. 1 boat carries 2 Missionaries must never be outnumbered by cannibals. Give a plan for all to cross the river.

   (b) Suppose you have the search space given in Table 1: [5]

State	Next	Cost
A	B	4
A	C	1
B	D	3
B	E	8
C	C	0
C	D	2
C	F	6
D	C	2
D	E	4
E	G	2
F	G	8

State	h
A	8
B	8
C	6
D	5
E	1
F	4
G	0

Table 1: Search space

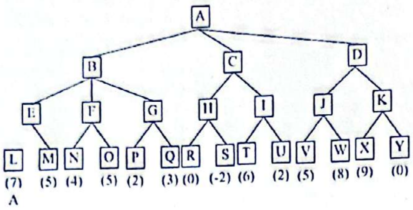
   i. Draw the state space of this problem.[0.5]

   ii. Assume that the initial state is A and the goal state is G. Show how each of the following search strategies would create a search tree to find a path from the initial state to the goal state. At each step of the following search algorithms, show which node is being expanded, and the content of fringe. Search algorithms: Uniform cost, Greedy search, A^* search [4.5]

   (c) Illustrate with an appropriate example how the Depth First Iterative Deepening Search Algorithm synthesizes the benefits of both Breadth First Search (BFS) and Depth First Search (DFS) algorithms, thereby achieving optimality, completeness, and linear space complexity. Nevertheless, the time complexity deteriorates in comparison to both BFS and DFS. [2]

3. (a) What distinguishes deterministic optimization algorithms from their non-deterministic counterparts? Do you consider that hill-climbing serves as a representation of a deterministic optimization algorithm? Please provide a rationale for your response. Examine the three limitations inherent in hill-climbing (HC) algorithms. Do you regard Simulated Annealing (SA) as an instance of a non-deterministic optimization algorithm? Please illustrate how SA addresses the limitations of HC. [4]

(b) Consider the following game tree in which static scores are all from the first players point of view: [4]



By analyzing this game tree, illustrate how the alpha-beta pruning algorithm minimizes the time complexity of Min-Max algorithm by applying the concept of pruning.

(c) What is CSP? Do you consider the following problem is an example of CSP? If so, identify the variables, domains and constraints associated with this problem. [1]
You have to color a planar map using only four colors, in such a way that no two adjacent regions have the same color.

4. (a) Define the following actions of a self-driving car using First-Order Logic (FOL). It is essential to declare the predicates before defining the actions. [5]
   i. The car can move to a destination if the destination is reachable and there are no obstacles on the road.
   ii. A self-driving car can proceed through an intersection if the traffic light is green and there are no pedestrians crossing.
   iii. How would you represent in FOL that a self-driving car should stop if there is an obstacle on the road?
   iv. A self-driving car should reduce its speed when entering a school zone if the school zone is active (e.g., during certain hours of the day).
   v. Represent in FOL the rule that a self-driving car can only change lanes if the adjacent lane is clear and the car is moving.

   (b) A drone is responsible for delivering a package to a destination. The drone starts at location L_0, and air route $R1$ connects L_0 to L_1, $R2$ connects L_1 to L_2, and $R3$ connects L_2 to the destination L_D. There is an obstacle (e.g., restricted airspace or a storm) on air route $R2$, but no obstacles on $R1$ or $R3$. Define the predicates, knowledge base, and rules in FOL, then use forward chaining to determine if the drone can reach the destination. [4]

Section B

5. (a) What is STRIPS? What is the Action Schema of STRIPS? Devise a planning problem employing STRIPS, concentrating on the delineation of preconditions and effects linked to the designated action. One may consider the following cargo handling challenges faced by an airport or select an alternative issue of your interest. [4]
   `Air Cargo Transport`- An air cargo transport problem involving loading cargo onto planes, unloading cargo off of planes, and flying the planes from one place to another.

   (b) Consider the following block-world problem: [5]

Start:	Goal:
ON (B,A)^	ON(C,A) ^
ONTABLE(A) ^	ON(B,D) ^
ONTABLE(C) ^	ONTABLE(A) ^
ONTABLE(D) ^	ONTABLE(D)
ARMEMPTY	

Solve this using partial order planning.

6. (a) Define forward and backward chaining. What are the factors that determine whether it is better to reason by using forward or backward chaining? [2]

   (b) What is conflict resolution? Discuss any two of the approaches which are used to solve conflict resolution. [1]

   (c) Draw the architecture of a rule-based system. Following are the rules and initial facts in the rule base and database of facts of a rule-based expert system. [6]

R1: IF A and C THEN E	Initial facts in the database
R2: IF C and D THEN F	A, B (both are true)
R3: IF B and E THEN F	Objective:
R4: IF B THEN C	Prove hypothesis G (goal)
R5: IF F THEN G	

Prove the goal (G) by applying both Forward Chaining and Backward Chaining inference procedures. You need to show the sequence of firing the rules and also need to draw the search tree, in which propagation of truth for FC and backtracking of goals for BC to be demonstrated diagrammatically.

7. (a) Answer the following questions by utilizing Bayes' Theorem. [3]
   i. A factory produces 60% of its products from Machine A and 40% from Machine B. The defect rate is 3% for Machine A and 5% for Machine B. If a randomly selected product is found to be defective, what is the probability that it was produced by Machine A?
   ii. A medical test for a certain disease has a 98% accuracy rate, meaning that 98% of people with the disease test positive and 98% of people without the disease test negative. If 1% of the population has the disease, what is the probability that a person who tests positive actually has the disease?
   iii. A spam filter is trained to classify emails as spam or not spam. The prior probability of an email being spam is 20%. The probability of an email containing the word "offer" given that it is spam is 80%. The probability of an email containing the word "offer" given that it is not spam is 10%. Calculate the probability that an email is spam if it contains the word "offer".

   (b) In a small town, an alarm system is designed to ring in case of a fire or an earthquake. When the alarm rings, two individuals, Mr. X and Mr. Y, may call to report the incident. The relationships between variables are as follows: [6]
   - **Fire** and **Earthquake** influence the probability of the **Alarm** ringing.
   - The **Alarm** influences the probability of calls from **Mr. X** and **Mr. Y.**

   The probabilities for the Bayesian Network are as follows:

   Probability of Alarm given Fire and Earthquake

$P(\text{Alarm} \mid \text{Fire, Earthquake}) =$

Fire	Earthquake	P(Alarm)
Yes	*Yes*	0.95
Yes	*No*	0.90
No	*Yes*	0.60
No	*No*	0.01

   Probability of Mr. X and Mr. Y calling given the Alarm

$P(\text{Mr. X Calls} \mid \text{Alarm}) =$

Alarm	P(Mr. X Calls)
Yes	0.80
No	0.10

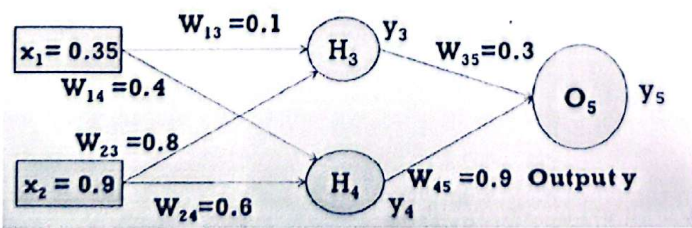
$P(\text{Mr. Y Calls} \mid \text{Alarm}) =$

Alarm	P(Mr. Y Calls)
Yes	0.70
No	0.20

   The probability of an Earthquake is 0.6, and the probability of a Fire is 0.4. Given the Bayesian Network and Conditional Probability Tables:
   i. Calculate probability of the event that the alarm has sounded but neither a fire nor an earthquake has occurred and both Mr. X and Y call.
   ii. Calculate probability of the event that the alarm has sounded and fire has occurred, an earthquake has not occurred and both Mr. X and Y call.

8. (a) What is associative memory? Is it your belief that neural networks are fundamentally rooted in the principles of associative memory? Please provide a more detailed explanation considering the structure and principles of multilayer neural network architecture. [3]

   (b) Consider the scenario where the neurons utilize a sigmoid activation function; perform both a forward pass and a backward pass through the network. The subsequent step involves performing the update of the weights. Let us consider that the true output of y is 0.6, with a learning rate established at 0.8. Execute an additional forward pass. [6]



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CSE 713: Artificial Intelligence
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[Answer any three questions from each of the following sections.]

SECTION A

A-1/ (a) What function does an intelligent agent serve? 4

(b) What primary component(s) will be used to evaluate the effectiveness of problem solving? 3

(c) What does a utility-based agent do? 2

A-2/ (a) What is resolution? Write down the algorithm for the conversion of well-formed formulas (wff) in predicate logic to clause form. 2.5

(b) Consider the following facts:

(F-1) Marcus was a man.
(F-2) Marcus was a Pompeian.
(F-3) Marcus was born in 40 A.D.
(F-4) All men are mortal.
(F-5) All Pompeian died when the volcano erupted in 79 A.D.
(F-6) No mortal lives longer than 150 years.
(F-7) It is now 2023.
(F-8) Alive means not dead.
(F-9) If someone dies, then he is dead at all later times.

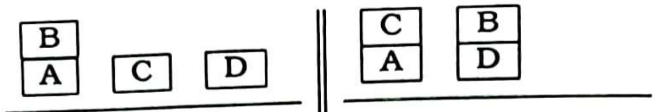
i. Translate these facts into well-formed formulas (wffs) in predicate logic. 1
ii. Answer the question *Is Marcus alive now?* using backward reasoning. 1
iii. Convert the formula into clause form. 1
iv. Prove that *Marcus is not alive now* using resolution. 2

(c) What do you mean by Knowledge Representation and Mapping? What are the roles of Knowledge Representation? 1.5

A-3/ (a) Write down the preconditions of the following operators: 2

i. UNSTACK(A,B)
ii. STACK(A,B)
iii. PICKUP(A)
iv. PUTDOWN(A)

(b) What is the goal stack planning? Consider the following block world problem: 5



Start:
ON(B, A)^
ONTABLE(A)^
ONTABLE(C)^
ONTABLE(D)^
ARMEMPTY

Goal:
ON(C, A)^
ON(B, D)^
ONTABLE(A)^
ONTABLE(D)

Solve this problem using goal stack planning/partial order planning/ STRIPS approach.

(c) Write short notes on Generative AI. 2

A-4/ (a) What is an AI Technique? Write down the advantages and disadvantages of Depth-First Search and Breadth-First Search. 2

(b) Describe the minimax search procedure. 4

(c) Explain the following with figures. 2

i. α-cutoff ii. β-cutoff iii. Futility-cutoff

(d) What do you mean by "*Waiting for Quiescence*" ? 1

SECTION B

B-1/ (a) What is Artificial Neural Networks? Compare artificial and biological networks. What aspects of biological networks are not mimicked by artificial ones? What aspects are similar? 3

(b) What is learning? Describe the inductive learning method with necessary example. 2

(c) What are the supervised learning, unsupervised learning, and reinforcement learning? 2

(d) Write short notes on the following: 2

i. Fuzzy logic ii. Uncertainty

B-2/ (a) In competitive learning networks, how are units at the input layer related to those at the second layer? 3

(b) Which layer in competing neural networks contains feedback weights? 3

(c) What prerequisites must exist before a competitive network can cluster patterns? 3

B-3/ (a) What is conflict resolution? Discuss any two of the approaches to the problem of conflict resolution. 2

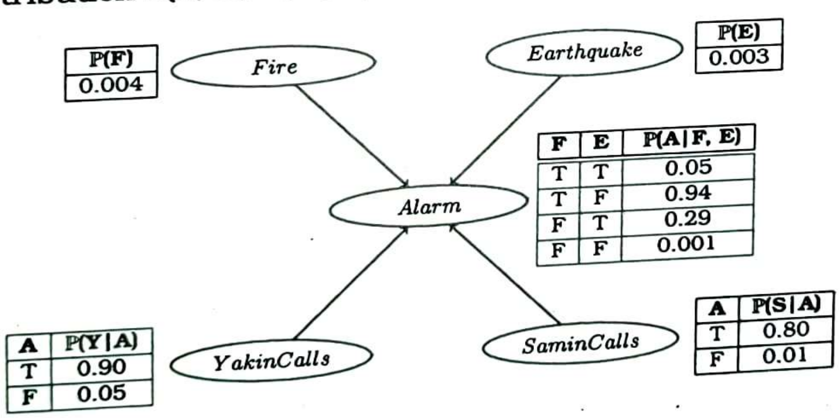
(b) Define forward and backward reasoning / chaining. What are the factors that determine whether it is better to reason forward or backward? 1.5

(c) Following are the rules and initial facts in the rule-base and database of facts of a rule-based expert system. 2

R1: if A and C then E	`Initial facts in the database`
R2: if C and D then F	`A, B (both are true)`
R3: if B and E then F	`Objective`
R4: if B then C	`Prove hypothesis G (goal)`
R5: if F then G	

Prove the goal (G) by applying both Forward Chaining and Backward Chaining inference procedures. You need to show the sequences of firing the rules and also need to draw the search tree, in which propagation of truth for FC and backtracking of goals fro BC to be demonstrated diagrammatically.

(d) What is Baye's Theorem? A Bayesian network (Figure 1), showing both the topology and the conditional probability tables (CPTs). In the CPTs, the letters **F**, **E**, **A**, **Y**, and **S** stand for *Fire*, *Earthquake*, **Yakin Callls**, **Samin Calls** respectively. The Independent conditional probability help us to write in a simplified way the joint distribution $\mathbb{P}$**(F, E, A, Y, S)**. 0.5



$\mathbb{P}$(F)
0.004

$\mathbb{P}$(E)
0.003

F	E	$\mathbb{P}$(A|F, E)
T	T	0.05
T	F	0.94
F	T	0.29
F	F	0.001

A	$\mathbb{P}$(Y|A)
T	0.90
F	0.05

A	$\mathbb{P}$(S|A)
T	0.80
F	0.01

Figure 1: Bayesian Network

i. Express the joint distribution $\mathbb{P}$**(F, E, A, Y, S)** in terms of the conditional probabilities (and independencies) expressed in the Bayesian Network above. 1

ii. Probability of the event that the alarm has sounded but neither a Fire nor an earthquake has occurred and both Yakin and Samin call. 1

iii. Probability of the event that the alarm has sounded and fire has occurred , an earthquake has not occurred and both Yakin and Samin call. 1

B-4/ (a) Write down **Steepest-Ascent Hill Climbing** algorithm. What are the problems that the hill climbing search process may reach? How the problems can be dealt? 2

(b) Suppose you have the following search space: 3

State	Next	Cost
A	B	4
A	C	1
B	D	3
B	E	8
C	C	0
C	D	2
C	F	6
D	C	2
D	E	4
E	G	2
F	G	8

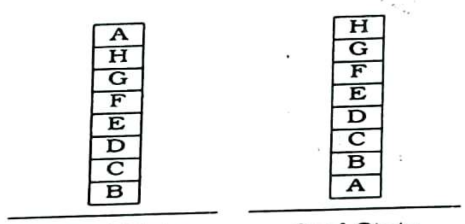
State	h
A	8
B	8
C	6
D	5
E	1
F	4
G	0

i. Draw the state space of this problem.

ii. Assume that the initial state is **A** and the goal state is **G**. Show how each of the following search strategies would create a search tree to find a path from the initial state to the goal state.

- Uniform cost
- Greedy search
- A* search

(c) What is heuristic function? Consider the following block world problem: 1



Initial State Goal State

Show that the hill climbing procedure is failed with local heuristic function but works perfectly with Global heuristic function.

(d) Trace the constraint satisfaction procedure solving the following cryptarithmetic problem: 3

```
    S E N D
+   M O R E
-----------
  M O N E Y
```

Initial State: `No two letters have the same value.` The sums of the digits must be as shown in the problem.

University of Chittagong

Department of Computer Science and Engineering

7th Semester B.Sc. (Engg.) Examination-2021

Course Code: CSE-713 Course Title: Artificial Intelligence

Total marks: 52.5 Marks Time: 4.00 hours

[Answer any ***three*** questions from each of ***Section-A*** and ***Section-B***. A separate answer script must be used for Section-A and Section-B. Figures in the right-hand margin indicate full marks.]

Section-A

1. a) Do you think humanoid Sophia and Jarvis possess the features and signs of intelligence? If so, elaborate your answer by taking into account the behavior of Sophia and Jarvis. 1.5

   b) Suppose you design a machine to pass the Turing test. What are the capabilities such a machine must have? 1

   c) Describe how Simple Reflex agents differ from Goal Driven agents by taking account of their architecture. 1

   d) Consider an intelligent agent named "*Amazon Echo*". Now, answer the following questions: 2+2

      i) Write a PAGE (Percepts, Actions, Goals, and Environments) description of the above intelligent agent.

      ii) Characterize the agent's environment with a proper explanation as being either: accessible or inaccessible; deterministic or non-deterministic; episodic or non-episodic; static or dynamic; and discrete or continuous.

   e) What agent architecture is best for the automated taxi driving agent? Justify your answer. 1.25

2. a) What is the state space search problem? 0.75

   b) Give the initial state, goal test, successor function, and cost function for each of the following. In addition, you need to elaborate the working principle of the basic search algorithm in each of the cases. 3

      i) Using only four colors, you have to color a planar map in such a way that no two adjacent regions have the same color.

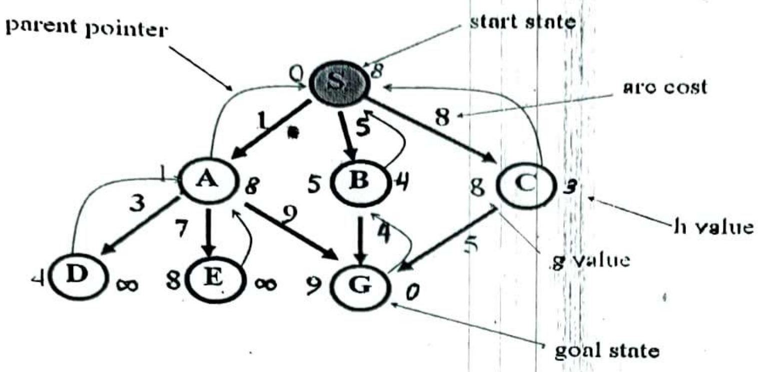
      ii) A 3-foot-tall monkey is in a room where some bananas are suspended from the 8-foot ceiling. He would like to get the bananas. The room contains two stackable, movable, climbable 3-foot-high crates.

   c) Suppose you have the following search space as shown in Figure 1: 3+2

      i) Assume that the initial state is S and the goal state is G. Show how each of the following search strategies would create a search tree to find a path from the initial state to the goal state.

         - Depth First Iterative Deepening
         - Greedy search
         - A* search

ii) At each step of the search algorithm, show which node is being expanded, and the content of the fringe. Also, report the eventual algorithm, and the solution cost.



N	h*(n)
S	9
A	9
B	4
C	5
D	∞
E	∞
G	0

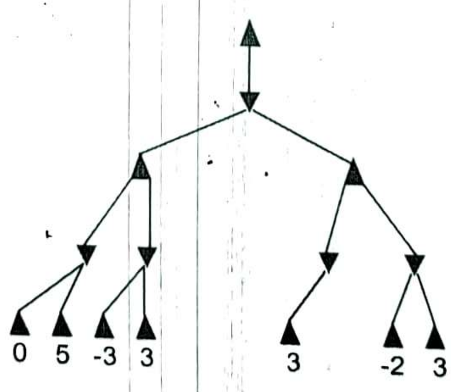
Figure 1

3. a) Write down Steepest-Ascent Hill Climbing algorithm. What are the problems that the hill-climbing search process may reach? How the problems can be dealt with? 1.5

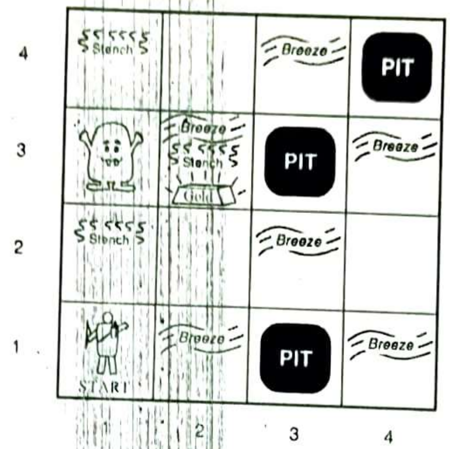
b) Differentiate between DFID and IDA* algorithms. Demonstrate how DFID incorporates the advantages of the Breadth First Search algorithm and the Depth First Search algorithm with an example. 1.75

c) What is CSP? Do you consider the following problem is an example of CSP? If so, identify the variables, domains, and constraints associated with this problem. 1.5
"*You have to color a planar map using only four colors, in such a way that no two adjacent regions have the same color.*"

d) Explain how does Alpha-Beta pruning improve the situation in the game playing in the following example. 4



4. a) The figure below shows a Wumpus World game, where the agent starts from location [1,1]. Briefly discuss with diagrams how the agent generates new knowledge through perceiving the environment and choosing his next move. 3



b) Differentiate between proof and theorem in one point? 0.75

c) Determine whether the following sentence is (i) Satisfiable, (ii) Contradictory, or (iii) Valid 2

$$(P \vee Q) \wedge (P \vee \sim Q) \vee P$$

d) Given knowledge base R1, R2, R3, R4, and R5 below, show that P1, 2 is false. 3.5

R1: $\sim$ P11

R2: B11 $\Leftrightarrow$ (P2,1 $\vee$ P1,2)

R3: B2,1 $\Leftrightarrow$ (P1,1 $\vee$ P2,2 $\vee$ P3,1)

R4: $\sim$ B1,1

R4: $\sim$ B1,1

R5: B2,1

Section-B

5. a) Briefly explain why FOL is considered as the generalization of PL. 1

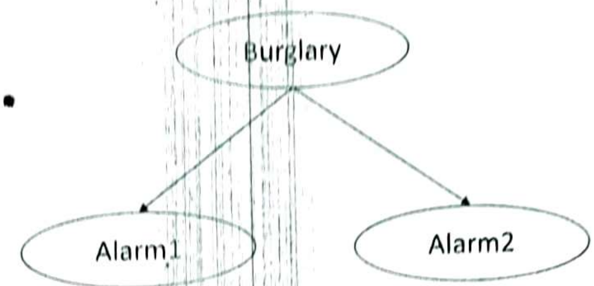
b) Consider the following sentences: 1.75+2.5+3.5

i) If a perfect square is divisible by a prime P, then it is also divisible by the square of P.
ii) Every perfect square is divisible by some prime,
iii) 36 is a perfect square,
iv) Does there exist a prime q such that the square of q divide 36?

Prove via the resolution method and unification (where needed) that "*Does there exist a prime q such that square of q divide 36?*" More precisely

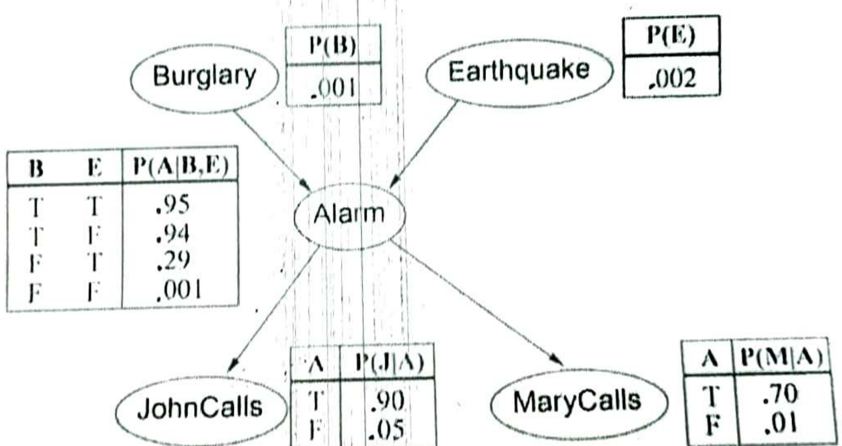
i) First, translate the sentences above into FOL sentences.
ii) Convert to Clause forms.
iii) Apply the resolution method with the unification (where needed) to prove the goal sentence: "*Does there exist a prime q such that the square of q divide 36?*"

8. a) To safeguard your house, you recently installed two different alarm systems by two different reputable manufacturers that use completely different sensors for their alarm systems. Considering these facts the Bayesian network is given below: 3.5



How many probabilities need to be specified for its conditional probability tables? How many probabilities would need to be given if the same joint probability distribution were specified in a joint probability table?

b) A Bayesian network (Figure 1), showing both the topology and the conditional probability tables (CPTs). In the CPTs, the letters B, E, A, J, and M stand for Burglary, Earthquake, Alarm John Calls, and MaryCalls, respectively. The Independent conditional probability helps us to write in a simplified way the joint distribution P(B, E, A, J, M). 5.25



i) Express the joint distribution P(F, E, A, Y, S) in terms of the conditional probabilities (and independencies) expressed in the Bayesian Network above.

ii) Probability of the event that the alarm has sounded but neither a Burglary nor an earthquake has occurred and both Mary and Jhon call.

iii) Probability of the event that the alarm has sounded and Burglary has occurred. an earthquake has not occurred, and both Mary and Jhon call.

6. a) Following are the rules and initial facts in the rule-base and database of facts of a rule-based expert system.

R1: IF A and C THEN E

R2: IF C and D THEN F

R3: IF B and E THEN F

R4: IF B THEN C

R5: IF F THEN G

Initial facts in the database

A, B (both are true)

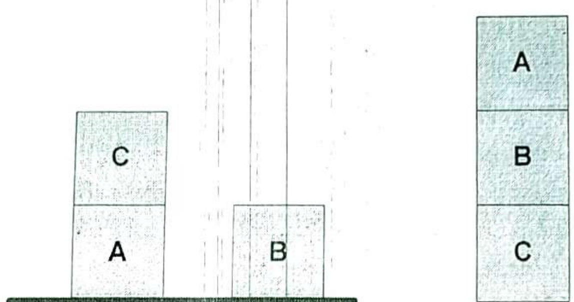
Objective:

Prove hypothesis G (goal)

Prove the goal (G) by applying both forward chaining (FC) and backward chaining (BC) inference procedures. You need to show the sequence of firing the rules and also need to draw the search tree, in which propagation of truth for FC and backtracking of goals for BC are to be demonstrated diagrammatically.

b) Consider the following blocks world problem known as Sussman Anamoly problem. Develop an effective and complete plan using Partial-Order Planning/STRIPS approach to convert the given initial state into goal state. It could be fine if you address the flaws such as. threats to causal links and open conditions in developing the final plan. 4

Sussman Anamoly Problem



Initial State Goal State

7. a) Taking into account of the example given below, explain the concept of uncertainty. 2.75

"The doorbell rang at 12'0 clock in the midnight.
Was someone there at the door?
Did Karim wake up?"

b) Explain why deductive and adductive reasoning are not considered as the examples of sound reasoning and hence, they are considered as the source of uncertainty. 2.5

c) What is Baye's theorem? A doctor knows that the disease meningitis causes the patient to have a stiff neck, say, 40% of the time. The doctor also knows some unconditional facts: the prior probability that a patient has meningitis is 1/50000, and the prior probability that any patient has a stiff neck is 1/25. Find the probability of patients with a stiff neck having meningitis. 3.5

University of Chittagong

Department of Computer Science and Engineering

7th Semester B.Sc. (Engg.) Examination-2020

Course Code: CSE-713 Course Title: Artificial Intelligence

Total marks: 52.5 Marks Time: 4.00 hours

[Answer any ***three*** questions from each of the ***Group-A*** and ***Group-B***. A separate answer script must be used for Group-A and Group-B. Figures in the right-hand margin indicate full marks.]

Group-A

1. a) Give a scientific definition of intelligence. Do you think the following definition of intelligence constitutes a scientific definition? If not, justify your answer. 2

"Intelligence is the acquiring and applying knowledge."

b) Discuss the different approaches of AI. Which one do you think is appropriate and why? 2

c) Find out the following about the automated taxi driving agent. 1+1.75+2

i) Identify its PAGE (Percepts, Actions, Goals, and Environment) description.

ii) Characterize its environment as being either: accessible or inaccessible; deterministic or nondeterministic; episodic or non-episodic; static or dynamic; and discrete or continuous; explaining what each of your selected terms means.

iii) What agent architecture is best for the Mars Rover? Justify your answer.

2. a) How can you formulate a search problem? Give examples of data structures that can be used to represent a state-space both in explicit and implicit ways. 1.75

b) Give the initial state, goal test, successor function, and cost function for each of the following. 2

i) In the travelling salesperson problem (TSP) there is a map involving N cities some of which are connected by roads. The aim is to find the shortest tour that starts from a city, visits all the cities exactly once, and comes back to the starting city.

ii) Missionaries and Cannibals problem: 3 missionaries and 3 Cannibals are on one side of the river. 1 boat carries 2 Missionaries must never be outnumbered by cannibals. Give a plan for all to cross the river.

c) Suppose you have the following search space: 1+3+1

State	Next	Cost
A	B	4
A	C	1
B	D	3
B	E	8
C	C	0
C	D	2
C	F	6
D	C	2
D	E	4
E	G	2
F	G	8

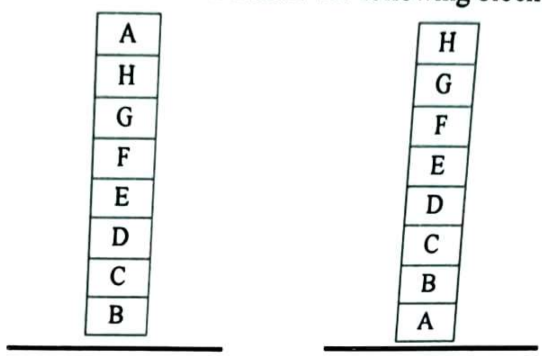
State	h
A	8
B	8
C	6
D	5
E	1
F	4
G	0

i) Draw the state-space of this problem.

ii) Assume that the initial state is A and the goal state is G. Show how each of the following search strategies would create a search tree to find a path from the initial state to the goal state. • Uniform cost • Greedy search • A* search

iii) At each step of the search algorithm, show which node is being expanded and the content of fringe.

3. a) What is a heuristic function? Consider the following block world problem:



Initial State Goal State

Fig.: A Hill Climbing Problem.

Show that the hill-climbing procedure is failed with the local heuristic function but works perfectly with the global heuristic function.

b) Iterative-deepening (ID) is a type of search algorithm, which is said to combine the benefits of breadth-first and depth-first search. Explain, using a diagram or otherwise, how ID achieves these benefits. 2

c) When is breadth-first search an *admissible* search strategy? Briefly explain. 1.75

d) Describe the Minimax algorithm for searching game trees. 2.5

4. Consider the following facts: 2+2+2+2.75

(i) Marcus was a man.
(ii) Marcus was a Pompeian.
(iii) Marcus was born in 40 A.D.
(iv) All men are mortal.
(v) All Pompeians died when the volcano erupted in 79 A.D.
(vi) No mortal lives longer than 150 years.
(vii) It is now 2021.
(viii) Alive means not dead.
(ix) If someone dies, then he is dead at all later times.

a) Translate these facts into well-formed formulas (WFFs) in predicate logic.
b) Answer the question "*Is Marcus alive now?*" using backward reasoning.
c) Convert the formulas into clause form.
d) Prove that "*Marcus is not alive now.*" using resolution.

Group-B

5. a) Define forward and backward chaining. What are the factors that determine whether it is better to reason by using forward or backward chaining? 1.5

b) What is conflict resolution? Discuss any two of the approaches, which are used to solve conflict resolution. 1.5

c) Draw the architecture of a rule-based system. Following are the rules and initial facts in the rule-base and database of facts of a rule-based expert system. 5.75

R1: IF A and C THEN E	Initial facts in the database
R2: IF C and D THEN F	A, B (both are true)
R3: IF B and E THEN F	Objective:
R4: IF B THEN C	Prove hypothesis G (goal)
R5: IF F THEN G	

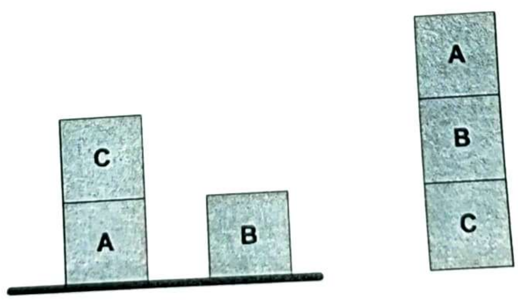
Prove the goal (G) by applying both forward chaining (FC) and backward chaining (BC) inference procedures. You need to show the sequence of firing the rules and also need to draw the search tree, in which propagation of truth for FC and backtracking of goals for BC are to be demonstrated diagrammatically.

6. a) Describe the STRIPS language for representing states, goals, and operators within a planning system. 2

b) Briefly explain why planning is difficult to cast as a state-space search problem. 1

c) Develop an effective and complete plan using the partial-order planning approach to convert the given initial state into the goal state. It could be fine if you address the flaws such as threats to causal links and open conditions in developing the final plan. 5.75

Sussman Anamoly Problem



Initial State Goal State

7. a) Taking into account of the example given below, explain the concept of uncertainty. 1.5

"The doorbell rang at 12'0 clock in the midnight.
Was someone there at the door?
Did Karim wake up?"

b) What will be happened if h' underestimates and overestimates h in A* algorithm? 2

c) What is an expert system? Describe the two essential capabilities of an expert system. 2

d) Why do we need the extension of the Bayes theorem? Write the equations and the semantics of the two extended Bayes theorems. Show two application areas of these extended Bayes theorems. Why do we need the extension of the Bayes theorem? 3

8. a) What is the Bayesian network? Describe the syntax and semantics of Bayesian networks with necessary examples. 1.5

b) A doctor knows that the disease meningitis causes the patient to have a stiff neck, say, 40% of the time. The doctor also knows some unconditional facts: the prior probability that a patient has meningitis is 1/50000, and the prior probability that any patient has a stiff neck is 1/25. Find the probability of patients with a stiff neck to have meningitis. 1.5

c) Your burglar alarm may or may not be sounding (node A). Possible causes of A are your house being burgled (node B) or a small earthquake (node E) setting it off by accident. Your reliable friend has called you saying that he thinks the alarm is sounding (node R). Your unreliable friend, who often calls when your alarm sounds, has not done so this time (node C). Your two friends do not have any contact with each other. 1+1+1+1+1.75

i) Draw a Bayesian network representing the variables, causes, and observations in this situation. The following are conditional probabilities for the events in the model:

$P(B) = 0.02$

$P(E) = 0.001$

$P(A|\neg B, \neg E) = 0.01$, $P(A|B, \neg E) = 0.9$, $P(A|\neg B, E) = 0.9$, $P(A|B, E) = 0.99$

$P(R|A) = 0.002$, $P(R|\neg A) = 0.999$ (NB these do not have to sum to 1)

$P(C|A) = 0.8$, $P(C|\neg A) = 0.1$ (NB these do not have to sum to 1)

ii) Compute the prior that the alarm is sounding.

iii) Compute the normalized likelihood that the alarm is sounding.

iv) Compute the posterior that the alarm is sounding.

v) Suppose that you hear a radio report that an earthquake has occurred near your home. Draw a modified Bayesian network to model this, and discuss qualitatively how your belief about each of A, E, and B would change. An earthquake has occurred and both Mary and Jhon call.

[redacted]

University of Chittagong
7^{th} Semester B.Sc (Engg.)Examination 2018
Department of Computer Science & Engineering
Course Code: CSE 713
Course Title: Artificial Intelligence

Time: 4 Hours **Full Marks:** $52\frac{1}{2}$

[Answer six questions taking any three from each section and use separate answer script for each section]

Section A

1. a) Define Intelligence. What are the different approaches in defining artificial intelligence? Which one do you think appropriate and why? 2
   b) Do you think humanoid *Sophia* and *Jarvis* possess the features and signs of intelligence? If so, elaborate your answer by taking account of the behavior of *Sophia* and *Jarvis*. 1.5
   c) What is intelligent behavior? Do you think only sensing and acting will make a machine intelligent? If not elaborate your answer. 1.5
   d) Find out the followings about Internet Shopping Agent.
      (i) Identify its PAGE description. 1.25
      (ii) Characterize its operating environment. 1.25
      (iii) How can one evaluate the performance of the agent? 1.25

2. a) Compare the basic search algorithms, which are used in both Tree and Graph data structures to represent a state space both in explicit and implicit ways. 1.5
   b) Give the initial state, goal test, sucessor function, and cost function for each of the following. 4
      i) A 3-foot-tall monkey is in a room where some bananas are suspended from the 8-foot ceiling. He would like to get the bananas. The room contains two stackable, movable,
      climbable 3-foot-high crates.
      ii You have three jugs measuring 12 gallons, 8 gallons, and 3 gallons, and a water faucet. You need to measure out exactly one gallon.
   c) Suppose you have the following search space:

State	Next	Cost
A	B	4
A	C	1
B	D	3
B	E	8
C	C	0
C	D	2
C	F	6
D	C	2
D	E	4
E	G	2
F	G	8

State	H
A	8
B	8
C	6
D	5
E	1
F	4
G	0

   i. Draw the state space of this problem. 1
   ii. Assume that the initial state is A and the goal state is G. Show how each of the following search strategies would create a search tree to find a path from the initial state to the goal state. 2.25
      Uniform cost
      Greedy search
      A* search

3. a) What is iterative improvement algorithm? Do you consider both Hill-Climbing and Simulated Annealing are of this branch of algorithm? If yes why? 2
   b) What is CSP? Do you consider the following problem is an example of CSP? If so, identify the variables, domains and constraints associated with this problem. 2
      You have to color a planar map using only four colors, in such a way that no two adjacent regions have the same color

c) Compare and contrast between DFID and IDA* algorithms. 1.75

d) Consider any three agents from this list: Medical Diagnosis System, Park-Picking Robot, Interactive English Tutor, Vacuum Cleaner, Internet Shopping and Mars Rover; then find out following for the three selected agents. 3

i. What are the percepts for this agent?
ii. Characterize its operating environment.
iii. What are the actions the agent can take?
iv. How can one evaluate the performance of the agent?

What sort of agent architecture do you think is most suitable for this agent?

4. a) Determine whether the following sentence is (i) Satisfiable, (ii) Contradictory (iii)Valid 3

$$(P \vee Q) \wedge (P \vee \sim Q) \vee P$$

b) Given knowledge base R1, R2, R3, R4, R5 below, show that P1, 2 is false. 3.75

R1: ~ P11
R2: B11 ⇔ (P2,1 ∨ P1,2)
R3: B2,1 ⇔ (P1,1 ∨ P2,2 ∨ P3,1)
R4: ~ B1,1
R5: B2,1

c) Prove FOL is considered as the generalization of PL. Explain with example why universal quantification (∀) elimination is easy while existential quantification (∃) elimination is not that much easy. 2

Section B

5. a)

i. If a perfect square is divisible by a prime **P,** then it is also divisible by square of **P**
ii. Every perfect square is divisible by some prime
iii. 36 is a perfect square
iv. Does there exist a prime **q** such that square of **q** divide 36?

Prove via the resolution method and unification (where needed) that "Does there exist a prime q such that square of q divide 36?" More precisely

a. First translate the sentences above into FOL Sentences 1.75
b. Convert to Clause Forms 3.5
c. Apply the resolution method with the unification (where needed) to prove the goal sentence: "Does there exist a prime **q** such that square of **q** divide 36?" 3.5

6. a) What is Rule-Based System (RBS)? Briefly describe the way in which IF-THEN rules can be used as a basis for knowledge representation and reasoning in RBS. 2

b) What is backtracking? Show how good conflict resolution strategy can be used to reduce backtracking. 2

c) Draw the architecture of a rule-based system. Following are the rules and initial facts in the rule-base and database of facts of a rule-based expert system. 4.75

R1: IF hot and Smokey THEN Add Fire	Initial facts in the database
R2: IF Alarm beeps THEN Add Smokey	Alarm beeps and hot (both are true)
R3: IF Fire THEN Add Switch Sprinkler	Objective: Prove hypothesis
	Switch Sprinkler (goal)

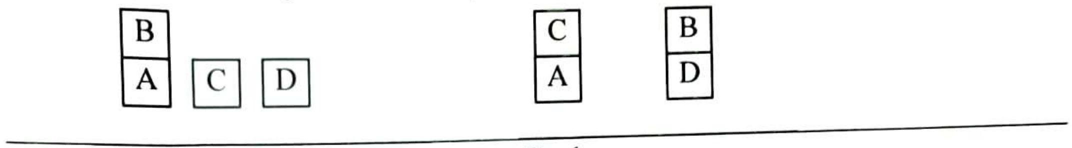
Prove the goal by applying both Forward Chaining and Backward Chaining inference procedures. You need to show the sequence of firing the rules and also need to draw the search tree, in which propagation of truth for FC and backtracking of goals for BC to be demonstrated diagrammatically.

7. a) What is planning? Write the factors, considered in formulating a planning problem. 1.5

b) Explain how planning differs from classical search techniques. 1.5

4.75

c) Consider the following blocks world problem. 5.75



Start:
ON(B,A)^
ONTABLE(A) ^
ONTABLE(C) ^
ONTABLE(D) ^
ARMEMPTY

Goal:
ON(C,A) ^
ON(B,D) ^
ONTABLE(A) ^
ONTABLE(D)

Solve this problem using goal stack planning/partial order planning/STRIPS approach.

8. a) Taking account of the example below, explain the concept of uncertainty. 1.5

" The doorbell rang at 12'0 clock in the midnight.
- Was someone there at the door?
Did Karim wakeup?"

b) Differentiate among adductive, deductive and default reasoning. Explain why they are not considered as sound reasoning and hence, considered as the sources of uncertainty. 3.25

c) A doctor knows that the disease meningitis causes the patient to have a stiff neck, say, 40% of time. The doctor also knows some unconditional facts: the prior probability that a patient has meningitis is 1/50000, and the prior probability that any patient has a stiff neck is 1/25. Find the probability of patients with a stiff neck to have meningitis. 4

University of Chittagong
Department of Computer Science and Engineering
7^{th} Sem. B. Sc. Engineering Examination, 2017
Course Code: CSE 713
Course Title: Artificial Intelligence
Total Marks: 52.5 Time: 4:00 Hours

[Answer any ***three*** questions from **Group-A** and any ***three*** questions from **Group-B;** Separate answer script must be used for Group-A and Group-B. Figures in the right-hand margin indicate full marks.]

Group-A

1. (a) What is search problem? Briefly explain why problem formulation must follow goal formulation. 1.25

(b) Give the initial state, goal test, operators, and path cost function for each of the following. 2.5

(i) Using only four colors, you have to color a planar map in such a way that no two adjacent regions have the same color.

(ii) A 3-foot-tall monkey is in a room where some bananas are suspended from the 8-foot ceiling. He would like to get the bananas. The room contains two stackable, movable, climbable 3-foot-high crates.

(c) Suppose you have the following search space:

State	Next	Cost
A	B	4
A	C	1
B	D	3
B	E	8
C	C	0
C	D	2
C	F	6
D	C	2
D	E	4
E	G	2
F	G	8

State	h
A	8
B	8
C	6
D	5
E	1
F	4
G	0

i) Draw the state space of this problem. 1
ii) Assume that the initial state is A and the goal state is G. Show how each of the following search strategies would create a search tree to find a path from the initial state to the goal state. 3

Uniform cost
Greedy search
A* search

(iii) At each step of the search algorithm, show which node is being expanded, and the content of fringe. Also report the eventual algorithm, and the solution cost. 1

2. (a) What are the major differences in resolution in PL and resolution in FOL. 1.25

(b) Consider the following sentences:

(1) Marcus was a man.
(2) Marcus was a Pompeian.
(3) Marcus was born in 40 A.D.
(4) All men are mortal.
(5) All Pompeian died when the volcano erupted in 79 A.D.
(6) No mortal lives longer than 150 years.
(7) It is now 2014.
(8) Alive means not dead.

i) Translate these facts into well-formed formulas (wffs) in predicate logic. 2

ii Convert the formulas into clause form. 2

iii) Prove that "Marcus is not alive now" using resolution. 2.5

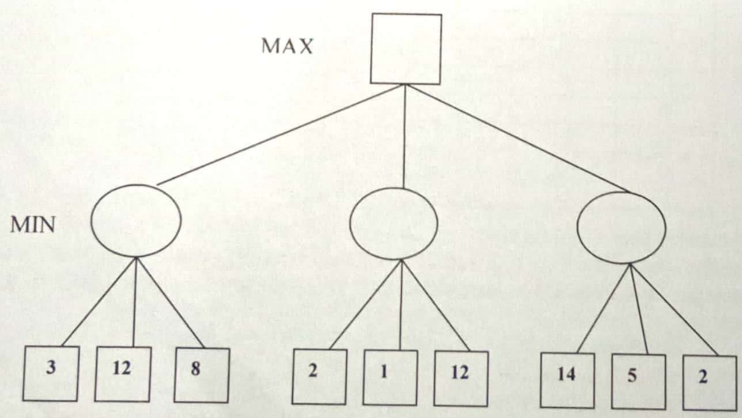
(c) Explain the concept of entailment with an appropriate example? 1

3. (a) Prove the optimality of A*. 2

(b) Is hill-climbing guaranteed to find a solution to the n-queens problem? Elaborate your answer. 2

(c) What is heuristic? Write the name of two heuristics associated with the eight puzzle problem. 1.5

(d) What is the Alpha-Beta pruning? How does pruning improves the situation in game playing? Elaborate your answer by taking account of the following tree. 3.25



4. (a) What is Bayesian Network? Describe the syntax and semantics of Bayesian Networks with necessary examples. 1.75

(b) Write the different types of inference procedures in BN. 1

(c) Your burglar alarm may or may not be sounding (node A). Possible causes of A are your house being burgled (node B) or a small earthquake (node E) setting it off by accident. Your reliable friend has called you saying that he thinks the alarm is sounding (node R). Your unreliable friend, who often calls when your alarm sounds, has not done so this time (node C). Your two friends do not have any contact with each other.

(i) Draw a Bayesian network representing the variables, causes and observations in this situation. 1

The following are conditional probabilities for the events in the model:

$P(B) = 0.02$

$P(E) = 0.001$

$P(A|\neg B,\neg E)=0.01$, $P(A|B,\neg E)=0.9$, $P(A|\neg B,E)=0.9$, $P(A|B,E)=0.99$

$P(R|A)=0.002$, $P(R|\neg A)=0.999$ (NB these do not have to sum to 1)

$P(C|A)=0.8$, $P(C|\neg A)=0.1$ (NB these do not have to sum to 1)

(ii) Compute the prior that the alarm is sounding. 1

(iii) Compute the normalised likelihood that alarm is sounding. 1

(iv) Compute the posterior that the alarm is sounding. 1

(v) Suppose that you hear a radio report that an earthquake has occurred near your home. Draw a modified Bayesian network to model this, and discuss qualitatively how your belief about each of A, E and B would change, earthquake has occurred and both Mary and Jhon call. 2

Section B

5. (a) Give a scientific definition of intelligence. Show how does it differ from dictionary definition of intelligence. 1

(b) "AI is the study and construction of rational agents" – do you agree with this statement? If yes, justify your answer. 1

(c) Suppose you design a machine to pass the Turing test. What are the capabilities such a machine must have? 1

(d) Find out the followings about the Automated Taxi Driving Agent..

(i) Identify its PAGE (Percepts, Actions, Goals and Environment) description. 0.75

(ii) Characterise it's environment as being either: accessible or inaccessible; deterministic or nondeterministic, episodic or non-episodic, static or dynamic, and, discrete or continuous; explaining what each of your selected term means. 3

(iii) What agent architecture is best for the Automated Taxi Driving Agent? Justify your answer. 2

6. (a) Define forward and backward chaining. What are the factors that determine whether it is better to reason forward or backward? 2

(b) What is conflict resolution? Discuss any two of the approaches, which are used solve conflict resolution. 2

(c) Draw the architecture of a rule-based system. Following are the rules and initial facts in the rule-base and database of facts of a rule-based expert system. 4.75

R1: IF A and C THEN E
R2: IF C and D THEN F
R3: IF B and E THEN F
R4: IF B THEN C
R5: IF F THEN G

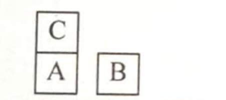
Initial facts in the database
A,B (both are true)
Objective:
Prove hypothesis G (goal)

Prove the goal (G) by applying both Forward Chaining and Backward Chaining inference procedures. You need to show the sequence of firing the rules and also need to draw the search tree, in which propagation of truth for FC and backtracking of goals for BC to be demonstrated diagrammatically.

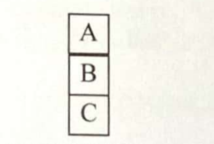
7. (a) What is the goal stack planning? Write down the specification of the following operators / actions:- 2.75

(i) UNSTACK (A,B), (ii) STACK(A,B), (iii) PICKUP(A), and (iv) PUTDOWN(A)

(b) Consider the following blocks world problem known as Sussman Anamoly Problem. Develop an effective and complete plan using Partial-Order Planning/STRIPS approach to convert given Initial State into Goal State. It could be fine if you address the flaws such as threats to causal links and open conditions in developing the final plan. 6



Initial State:
ON(C, A) ^
ONTABLE(A) ^
ONTABLE(B) ^
ARMEMPTY



Goal State:
ON(B, C) ^
ON(A, B) ^
ONTABLE(C)

8. (a) What is Hypothesis? Describe how the concept of hypothesis is utilized in the inductive learning method. 2.75

(b) What is uncertainty? What are the different sources of uncertainty? 2

(c) Prove FOL is considered as the generalization of PL Briefly describe why FOL is more expressive than PL. 2

(d) Explain with example why universal quantification (∀) elimination is easy while existential quantification (∃) elimination is not that much easy 1

(e) Prove Modus Ponens is very much a sound inference procedure. 1

University of Chittagong

Department of Computer Science and Engineering
7^{th} Semester B.Sc. Engineering Examination, 2016
CSE713 Artificial Intelligence
Total Marks: 52.5 Time: 4:00 Hours

[Answer any *three* questions from **Group-A** and any *three* questions from **Group-B;** Separate answer script must be used for Group-A and Group-B. Figures in the right-hand margin indicate full marks.]

Group-A

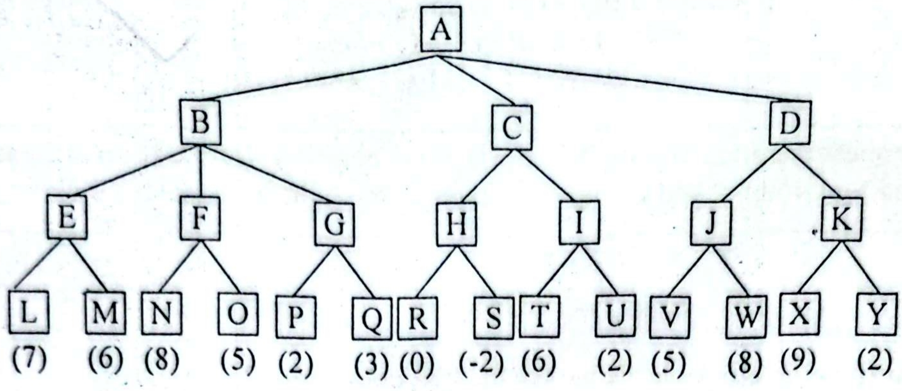
1. (a) What is Artificial Intelligence? What are the task domains of AI? 2
   (b) What is Intelligent Agent? Describe different agents with necessary block diagram. 3
   (c) How can you formulate a search problem? Give examples of data structures that can be used to represent a state space both in explicit and implicit ways. 3.75

2. (a) Write down the unification algorithm. 2
   (b) Explain with example why universal quantification (∀) elimination is easy while existential quantification (∃) elimination is not that much easy. 0.75
   (c) What is resolution? Write down the algorithm for conversion well-formed formulas (wffs) in predicate logic to clause form. 2
   (d) Consider the following facts:- 4
      (1) Marcus was a man.
      (2) Marcus was a Pompeian.
      (3) Marcus was born in 40 A.D.
      (4) All men are mortal.
      (5) All Pompeian died when the volcano erupted in 79 A.D.
      (6) No mortal lives longer than 150 years.
      (7) It is now 2014.
      (8) Alive means not dead.
      (9) If someone dies, then he is dead at all later times.
   (i) Translate these facts into well-formed formulas (wffs) in predicate logic.
   (ii) Answer the question "Is Marcus alive now?" using backward reasoning.
   (iii) Convert the formula into clause form.
   (iv) Prove that "Marcus is not alive now" using resolution.

3. (a) What do you understand by knowledge representations and mapping? Explain with necessary figure and example. 2
   (b) Write short notes on (i) simple relational knowledge, and (ii) inheritable knowledge. 2
   (c) Define forward and backward reasoning/chaining. What are the factors that determine whether it is better to reason forward or backward? 2.75
   (d) What is conflict resolution? Discuss any two of the approaches to the problem of conflict resolution. 2

4. (a) Describe the minimax search procedure. 2.75
   (b) What is the Alpha-Beta pruning? How does pruning improves the situation in game playing? Elaborate your answer with an example. 2

(c) Consider the following game tree in which static scores are all from the first player's point of view:



(i) Suppose the first player is the maximizing player. What move should be chosen?
(ii) In the game tree, what nodes would not need to be examined using the alpha-beta pruning procedure?

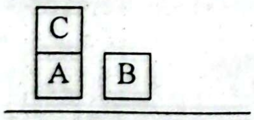
(d) Write short notes on (i) Futility- cutoff and (ii) Waiting for Quiescence. 2

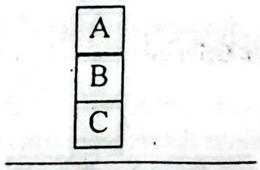
Group-B

5. (a) What is the goal stack planning/partial order planning? Write the factors, considered in formulating a planning problem. 2

(b) Write down the preconditions of the following operators:- 2
(i) UNSTACK (A,B), (ii) STACK(A,B), (iii) PICKUP(A) and (iv) PUTDOWN(A)

(c) Consider the following blocks world problem known as Sussman Anamoly Problem. 4.75
Develop an effective and complete plan using Partial-Order Planning/STRIPS approach to convert given Initial State into Goal State. It could be fine if you address the flaws such as threats to causal links and open conditions in developing the final plan.





Initial State:
ON(C, A) ^
ONTABLE(A) ^
ONTABLE(B) ^
ARMEMPTY

Goal State:
ON(B, C) ^
ON(A, B) ^
ONTABLE(C)

6. (a) What is learning? Describe the inductive learning method with necessary example. 2.25

(b) What is Artificial Neural Networks? Compare artificial and biological networks. What aspects of biological networks are not mimicked by artificial ones? What aspects are similar? 2

(c) Write short notes on the following : 2.5
(i) Fuzzy logic, and (ii) Uncertainty

(d) What are the supervised learning, unsupervised learning and reinforcement learning? 2

7. (a) What is Baye's theorem? A doctor knows that the disease meningitis causes the patient to have a *stiff neck*, say, 40% of time. The doctor also knows some unconditional facts: the *prior probability* that a patient has meningitis is 1/50000, and the prior probability that any patient has a *stiff neck* is 1/25. Find the probability of patients with a stiff neck to have meningitis. 1.5

(b) A Bayesian network (*Figure 1*), showing both the topology and the conditional probability tables (CPTs). In the CPTs, the letters F, E, A, Y and T stand for Fire, Earthquake, Alarm, Yakin Calls, and Tashfin Calls, respectively. The Independent conditional probability help us to write in a simplified way the joint distribution P(F,E,A,Y,T). 3

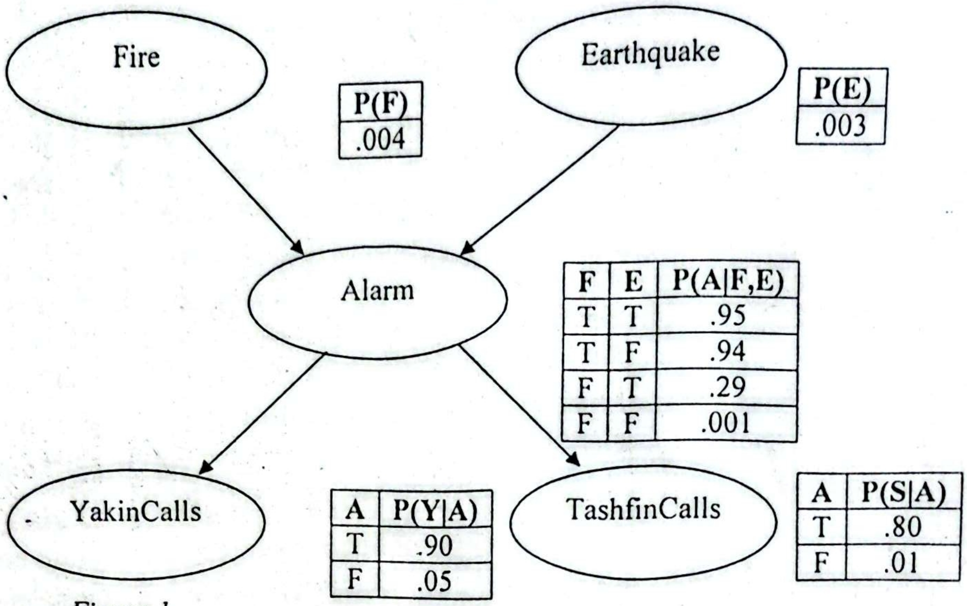


Figure 1

(i) Express the joint distribution P(F,E,A,Y,T) in terms of the conditional probabilities (and independencies) expressed in the Bayesian Network above.

(ii) Probability of the event that the alarm has sounded but neither a Fire nor an earthquake has occurred and both Yakin and Tashfin call.

(iii) Probability of the event that the alarm has sounded and fire has occurred, an earthquake has not occurred and both Yakin and Tashfin call.

(c) What is Bayesian Network? Describe the syntax and semantics of Bayesian Networks with necessary examples. 2

(d) What is heuristic function? Consider the following block world problem: 2.25

Fig: A Hill Climbing Problem

Show that the hill climbing procedure is failed with local heuristic function but works perfectly with Global heuristic function.

8. (a) What is Expert System? Briefly describe the major expert system components. 2

(b) What is Digital Signature? Briefly describe with necessary example. 2.5

(c) What is NLP? Write about the steps through which NLP is conducted. 2

(d) Write short notes on AND-OR graphs. Do you think best-first search algorithm is adequate for searching AND-OR graphs? 2.25